

From Natural-Language Hypotheses to Verified Data-Mining Discoveries

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Abstract—Fluent LLM hypotheses are useful data-mining candidates, but unsafe as discoveries: a generated claim may be unsupported, confounded, multiplicity-induced, or unstable. We study LLM-guided verifiable discovery: we let an LLM suggest constrained natural-language hypotheses, but we ask a separate verifier to decide which claims deserve to be called discoveries. VeraDM compiles temporal, sequence, subgroup-treatment, and conjunctive hypotheses into typed data-mining queries, then accepts them only after support checks, permutation testing, multiple-testing correction, matched-contrast validation, and held-out replication. Every accepted or rejected claim becomes an auditable evidence card. Under the split and firewall assumptions, we prove firewall-preservation and frozen-batch FDR results, give BY and multi-round variants, and show a power-separation result explaining why an informative proposer preserves power where exhaustive BH collapses. In experiments, synthetic failure-mode studies, semi-synthetic injections, statistically sound pattern-mining baselines, multi-provider proposer diagnostics, timestamped Bank/Bike/Healthcare case studies, and audit controls show that VeraDM rejects plausible invalid claims while producing readable, replicated discoveries.

Index Terms—statistically sound pattern mining, trustworthiness and interpretable data mining, large language models for data mining, mining time-evolving data, false discovery control, matched-contrast validation.

I. Introduction

Interpretable discovery is a central goal of data mining. Association rules, contrast sets, emerging patterns, subgroup descriptions, and rule lists [1], [2], [3], [4], [5] all seek structure that a human can inspect, question, and reuse. Their interpretability comes from a symbolic interface: a discovered pattern is a statement about a condition under which a target changes.

Large language models (LLMs) are a natural front end for this process. Given a schema, they can write hypotheses in plain language—for example, “customers with high debt and low savings are more likely to default”—that connect feature names, domain vocabulary, and candidate rule forms. Adjacent LLM-assisted analysis systems already show how useful this interface can be [6], [7]. The difficulty is that the interface is more fluent than it is trustworthy. A confident, grammatical claim may name nonexistent fields, describe under-supported patterns, or report associations that held only in the sample that inspired them [8]. Even

a syntactically valid claim can be an artifact of data dredging or distribution shift [9], [10], [11].

We therefore argue that LLM-guided data mining needs a strict separation between proposal and acceptance: the LLM proposes, but a verifier compiles each claim into a typed query and accepts it only after explicit evidence gates. This separation is the paper’s central design choice. It lets us use the model for what it does well, namely finding plausible and human-readable candidates, while leaving authority over what counts as a discovery to statistical tests, held-out evidence, and an auditable record.

We introduce VeraDM, a framework for LLM-guided data mining with verifiable discovery. Throughout, a “natural-language hypothesis” means a constrained grammar (Sec. III), not open-ended language understanding: the compiler fails closed on anything it cannot type-check. The LLM’s flexibility is in which typed claims to propose, and the verifier alone decides which claims are accepted. Concretely, on Bank Marketing, VeraDM accepts “long call duration and low euribor $\Rightarrow$ subscription” (validation effect .59, held-out .58) yet rejects the equally fluent “contacted by cellular $\Rightarrow$ subscription”, which shows no held-out effect, and it reports both as evidence cards that name the gate that decided them.

Contributions. (1) Protocol and compiler: we formulate LLM-guided verifiable discovery as proposal/acceptance separation and give a typed compiler from constrained natural-language hypotheses to executable pattern queries. (2) Verifier and statistical results: we combine support, permutation testing, BH/BY correction, matched contrast, and held-out replication, prove firewall-scoped FDR results, and analyze weighted and multi-round variants. (3) Benchmarks and evidence cards: we report synthetic failure regimes, real-covariate injections, timestamped case studies, proposer diagnostics, and auditable evidence cards in which every verdict names the gate that decided it.

Table I is our reader’s map: what each major claim is, where we support it, and what limitation remains.

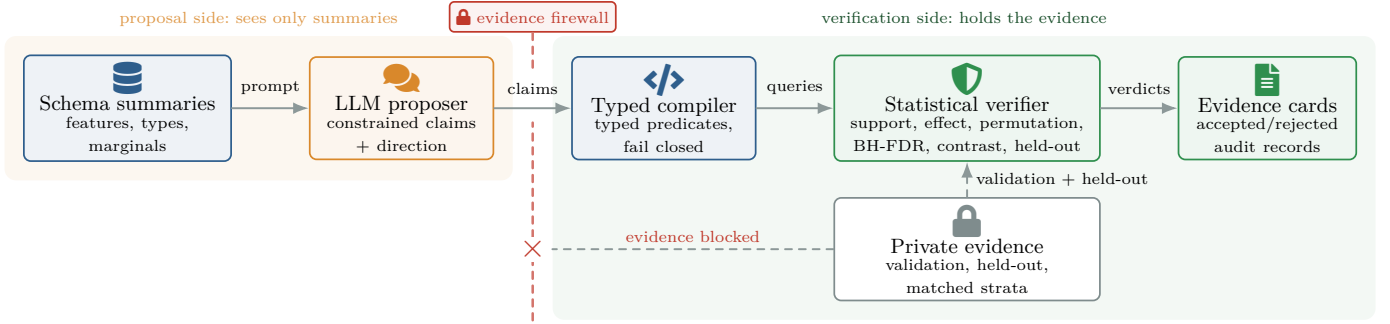


Fig. 1. VeraDM splits discovery into a proposal side and a verification side; validation and held-out evidence stay behind the firewall.

TABLE I
Claim-to-evidence map for the main argument.

Claim	Where	Key evidence	Limitation
proposal– acceptance split	Alg. 1, Fig. 1	LLM proposes, verifier accepts	constrained grammar
frozen-batch FDR	Thm. 1	89 prompts, 0 leaks; BY survival	public-data contamination risk
power separation	Thm. 3	Tables IV, V	depends on proposer ρ
matched contrast	Fig. 4	16/11/13 rejects fail it	exact-match overlap varies
matters evidence cards are auditable	Table XIII	every verdict names its gate	not user-validated

II. Related Work

Pattern and rule mining. Association rules [1], contrast sets [2], emerging patterns [3], subgroup discovery [4], [5], [12], [13], [14], [15], pattern-set mining [16], [17], and rule-list learning [18], [19] define symbolic pattern families and search them. VeraDM keeps that interpretable rule interface but changes how candidates enter the tested family: constrained natural language proposes a frozen batch, and a separate verifier decides acceptance.

Statistically sound mining and adaptive inference. BH-FDR [9], [20], permutation tests [21], Tarone-style testing [22], [23], Webb-style adjustments [24], [25], Kingfisher [26], LAMP-style mining [27], SPuManTE [28], and statistically sound pattern surveys [29] study enumerated families and correct over those search spaces, and we compare against representative approximations of this line (Tarone-style, Westfall–Young maxT-style, and Subgroup-BH) rather than re-implementing every system. We make no claim of superiority over LAMP or SPuManTE, and a direct empirical comparison with them is left to future work. VeraDM instead controls a language-proposed batch after it has been frozen behind a firewall. Selective and online inference [10], [30], [31], [32], [33] study repeated or adaptive testing, and sample splitting [34] and post-selection inference [35], [36] offer stronger frameworks. We adopt sample splitting because it stays valid for an arbitrarily adaptive language proposer, whereas post-selection inference would require a tractable description of the selection event. VeraDM therefore freezes the batch

first, then tests and replicates it on held-out data.

Robustness and LLM-assisted analysis. Invariant prediction and domain generalization seek stability across environments [37], [38], [39], and matching and counterfactual comparisons diagnose confounding [11], [40], [41]. VeraDM uses matched strata as a diagnostic gate, not as a causal identifier. LLM-assisted analysis and visualization systems [6], [7], [42], [43], [44], [45] make analysis easier but usually optimize for insight generation. VeraDM frames the LLM as a candidate generator for statistically validated pattern discovery. Our question is narrower and, for data mining, more consequential: after language proposes a claim, what evidence makes it safe to publish?

III. Problem Setup and the VeraDM Framework

Problem setup. Let $D = \{(x_i, y_i, s_i)\}_{i=1}^n$ have typed features x_i , a binary target y_i , and a split label $s_i \in \{\text{summ}, \text{val}, \text{hold}\}$. The proposer observes only a statistic $S(D_{\text{summ}})$ (schema, marginal summaries, discovery-set associations). A hypothesis h pairs a natural-language statement with a compiled query $q_h = (A_h, r_h, \eta_h)$, where A_h is a conjunction of predicates, $r_h \in \{-1, +1\}$ is the asserted direction, and η_h names the estimand kind. The association estimand is $\theta_h = \mathbb{P}(Y=1 \mid A_h=1) - \mathbb{P}(Y=1 \mid A_h=0)$, and a subgroup-treatment estimand instead extracts a treatment predicate T_h and subgroup G_h . We write $\Delta_{\text{val}}(h)$ and $\Delta_{\text{hold}}(h)$ for the plug-in estimates of θ_h on the validation and held-out splits. The verifier accepts h iff the signed effect $r_h \theta_h$ is large enough, statistically reliable, robust to matched contrast, and replicated on held-out data. The rest of the section turns that sentence into the concrete workflow of Fig. 1: what the proposer may see, what the compiler is allowed to emit, which gates decide acceptance, and what evidence card a reader receives at the end.

Proposal. We start deliberately upstream of testing. The proposal module receives a schema and (optionally) discovery-set summaries, and emits constrained hypotheses of the form IF <predicate> [AND ...] THEN $y=1$ increases/decreases, where the asserted direction is reported as $\uparrow$ (increase) or $\downarrow$ (decrease).

Compiler. We then remove ambiguity before any statistical decision is made. The compiler maps each hypothesis to a typed query and fails closed on any unknown feature, illegal operator, or missing direction, which is what makes the firewall claim of Proposition 1 checkable. It supports equality predicates, numeric thresholds, temporal windows (EVENT x WITHIN k DAYS $\Rightarrow$ event x within kd), ordered sequences (SEQUENCE $x \rightarrow y$ WITHIN k DAYS), and subgroup “treatment” clauses (TREATMENT x AMONG g) that we read association-ally rather than causally.

Gates. Once a claim is typed, we ask it to survive a sequence of increasingly strict checks rather than a single significance test. For each compiled query, VeraDM applies six gates. The support gate requires validation coverage of at least τ , checked before any held-out record is read; the held-out replication gate re-checks coverage on the held-out split. The validation effect gate requires $|\Delta_{\text{val}}(h)|$ to exceed ϵ . A one-sided permutation test runs R rounds on the validation split, using the signed effect as the test statistic and label permutation as the null. BH-FDR then corrects this validation p -value family over the entire frozen compiled batch, before any gate removes a candidate; held-out effect confidence intervals use a bootstrap. The matched-contrast gate compares exact-match context strata and requires the signed matched effect to exceed ϵ . Finally, held-out replication requires the signed held-out effect and its CI to support the asserted direction. Exact matching can be low-overlap, which would make a matched effect of zero misleading, so the gate also reports the fraction of mass retained in non-empty overlap strata and abstains when this falls below 0.5. Abstention is conservative rather than a soft warning: the card is reported as not certified instead of accepted, so a low-overlap rule never counts as a discovery. We surface this fraction for every accepted card (Table VII).

Treatment clauses. For a TREATMENT x AMONG g hypothesis, matched contrast is a diagnostic that screens confounding on the context Z_h (either LLM-suggested or a default set), not a causal identification strategy. We therefore report an accepted treatment card as a treatment-associated contrast: an association that survived matching on Z_h , not an identified causal effect. The default context is fixed in advance by the verifier rather than chosen per result: Z_h is the pair of leading schema features not already used by the rule’s own predicates, which gives {age $\geq$ 50, married} on Bank, {warm season, working day} on Bike, and {age $\geq$ 70, stay $\geq$ 7 days} on Healthcare. A two-step variant instead asks the proposer for Z_h and an identifying assumption (Alg. 1).

Evidence cards. The reader-facing output is a card rather than a standalone claim, because we want the verdict and the reasons for it to travel together. Each accepted or rejected card contains the constrained natural-language hypothesis, compiled query, support, validation effect, permutation p , adjusted q , matched contrast, held-

TABLE II
Evidence-card schema.

Field group	Contents
claim/query	constrained claim, typed predicate, asserted direction
validation	support, effect, permutation p , adjusted q , matched contrast
held-out	support, effect, bootstrap CI, BY flag when requested
decision	accepted/rejected, failed gates, redundancy-pruning note
audit	split identifiers, proposer/model metadata

Algorithm 1: VeraDM: LLM-Guided Verifiable Discovery

Input: dataset D , discovery summary $S(D_{\text{summ}})$, proposal model M , thresholds τ, ϵ, α

- 1 prompt M with $S(D_{\text{summ}})$ to obtain hypotheses H
- 2 optionally ask M for context variables Z_h and an identifying assumption for each h
- 3 compile each $h \in H$ into typed query
 $q_h = (A_h, r_h, \eta_h)$; reject failures
- 4 forall compiled h do
 - 5 estimate $\Delta_{\text{val}}(h)$, support, permutation p_h
 - 6 compute matched-strata contrast C_h using Z_h or defaults
 - 7 estimate $\Delta_{\text{hold}}(h)$ and its bootstrap CI
- 8 apply BH correction to the validation p -value family of the whole batch
- 9 accept h iff support, effect, batch FDR, contrast, and replication gates pass
- 10 prune accepted cards by coverage overlap and predicate supersets
- 11 return accepted and rejected evidence cards

out effect and CI, and rejection reasons (Table II). A redundancy-aware post-pruning step removes accepted cards whose validation coverage has Jaccard overlap > 0.9 with a stronger accepted rule, or that are strict predicate supersets of one.

IV. Theoretical Analysis

The formal results are meant to capture the discipline in Fig. 1. Under the stated split and firewall assumptions, we allow the LLM to adapt to discovery-set summaries, but not to the evidence used for final acceptance. This is the narrow condition that lets us combine flexible proposal with ordinary multiple-testing logic. Table III maps each result to the operational concern it addresses. Split the sample into mutually independent $D_{\text{summ}}, D_{\text{val}}, D_{\text{hold}}$. The proposer observes only $S(D_{\text{summ}})$ and external randomness independent of $(D_{\text{val}}, D_{\text{hold}})$, and emits a compiled batch $H_1, \dots, H_m$. Let p_i^{val} be a validation p -value for $H_{0,i} : r_i \theta_i \leq 0$.

Proposition 1 (Compiler soundness and firewall preservation): Every accepted compiler output $C(u) = q$ is

TABLE III
Theory map linking results to operational concerns.

Result	Main assumption	Reader payoff	Check
compiler firewall	fail-closed parser	blocks hidden evidence access	prompt audit
batch FDR	frozen batch; PRDS	BH valid on frozen family	null sweep
BY fallback	arbitrary dependence	conservative fallback	BY survival
weighted BH	pre-validation weights	optional power gain	Tab. XII
power separation	high proposer ρ	small batch preserves power	Tabs. IV, V
adaptive lower bound	reused verdicts/data	rules out naive interaction	multi-round
wealth spending	fresh shards; spent α_t	interaction with budget	multi-round

a well-typed query referring only to schema fields or predeclared engineered features. If u is generated from $S(D_{\text{summ}})$ and external randomness, then $C(u)$ is measurable with respect to the same information and independent of $(D_{\text{val}}, D_{\text{hold}})$ conditional on $S(D_{\text{summ}})$.

Proof sketch. The compiler performs deterministic parsing, schema lookup, type and direction checking, and it does not query validation or held-out records. Composition of the proposer with C cannot introduce dependence on $(D_{\text{val}}, D_{\text{hold}})$. $\square$

Theorem 1 (Frozen-batch FDR under an adaptive proposer): Assume (A1) $D_{\text{summ}}, D_{\text{val}}, D_{\text{hold}}$ are mutually independent, (A2) $(H_1, \dots, H_m)$ is measurable w.r.t. $S(D_{\text{summ}})$ and external randomness independent of $(D_{\text{val}}, D_{\text{hold}})$, (A3) each true-null validation p -value is super-uniform conditional on the proposed hypothesis, and (A4) the validation p -values satisfy PRDS on the null coordinates. Applying BH at level α to $\{p_i^{\text{val}}\}$ yields $\text{FDR} \leq |\mathcal{H}_0|/m \cdot \alpha \leq \alpha$, where $\mathcal{H}_0$ is the set of true nulls in the batch.

Proof sketch. Condition on the realized batch. By (A1)–(A2) the validation sample is independent of which H_i was proposed. Conditional on the batch, (A3)–(A4) are the BH conditions [9], [20]. Integrating gives the marginal bound. $\square$

Which family is corrected. The key bookkeeping point is simple but important: BH corrects the validation p -values of the entire frozen compiled batch $\{H_i\}_{i=1}^m$, fixed by the firewall before any validation or held-out record is touched, and it is applied once, to that whole family, before any gate removes a candidate. Every other check—support, validation effect, matched contrast, held-out support, held-out effect and CI direction, and redundancy pruning—acts afterward and can only remove cards, so the displayed set is a subset of the BH-controlled set. Because filtering after BH can in principle change the realized false-discovery proportion, we state Theorem 1 for the BH step and read the remaining gates as conservative, discovery-reducing checks rather than part of the FDR argument. The held-out split therefore carries replication rather than the FDR

guarantee itself. BH vs. BY in practice. We recommend BH as the default when proposed rules are approximately independent or PRDS, and BY (Corollary 1) when the batch contains many heavily overlapping predicates whose validation statistics are strongly co-dependent. In all three case studies the families overlap yet every BH accept also survives BY (Table X), so we report BH with BY as a routine robustness check rather than switching defaults. When dependence cannot be characterized at all, two drop-in alternatives avoid the PRDS question entirely: e-BH [46] controls FDR under arbitrary dependence without the BY correction factor, and conformal selection [47], [48] replaces a parametric interval with a distribution-free calibration. We do not use either here, but they are the principled fallbacks when a proposed batch is adversarially correlated.

Corollary 1 (Arbitrary dependence): If PRDS is not assumed, the Benjamini–Yekutieli correction (BH with critical values divided by $c_m = \sum_{k=1}^m 1/k$) controls FDR at level α under arbitrary dependence [20].

Theorem 2 (LLM-confidence weighted BH): For nonnegative weights $w_i = g(c_i)/\bar{g}$ built from the proposer’s stated confidence c_i , with $m^{-1} \sum_i w_i = 1$ and measurable w.r.t. $(S(D_{\text{summ}}), \text{rand. ind. of } D_{\text{val}})$, applying BH to weighted p -values p_i^{val}/w_i [33] controls FDR at level α under the conditions of Theorem 1.

Theorem 3 (Power separation from an informative proposal prior): Consider an expanded family of size M containing a non-null hypothesis with effect $\theta > 0$. Let an exhaustive verifier test all M claims by BH at level α , and let a firewall-safe proposer output $m \ll M$ claims that contain the non-null hypothesis with probability ρ independent of D_{hold} . Under standard sub-Gaussian one-sided tests, exhaustive testing requires $n \geq c_1 \theta^{-2} \log(M/\alpha)$ held-out samples for constant detection probability, while the proposed family requires $n \geq c_2 \theta^{-2} \log(m/\alpha)$ conditional on proposal. The unconditional proposed-family power is at least ρ times its conditional power.

Proof sketch. Passing BH in a family of size k needs a null tail of order α/k , so detecting θ with constant probability requires $n = \Omega(\theta^{-2} \log(k/\alpha))$. Substitute $k = M$ and $k = m$. The firewall ensures the proposed family does not invalidate the p -values. $\square$

Theorem 4 (No free FDR control under verdict-adaptive proposal): For any protocol that lets a proposer observe verdicts and then test a new nonempty batch at per-round level at least $\alpha_0 > 0$ without a finite alpha budget, there exists an all-null sequence with valid conditional p -values whose cumulative FDR after T rounds is at least $1 - (1 - \alpha_0)^T \rightarrow 1$.

Theorem 5 (Validation-only BH can be anti-conservative): If the proposer can observe a nonconstant statistic of the split used to compute the BH-controlled p -values, there exists a null distribution and a proposer for which BH at any fixed α has $\text{FDR} \rightarrow 1$ as the number of candidate features grows.

This is the failure mode the firewall rules out: the proposer sees only $S(D_{\text{summ}})$, which is independent of D_{val} by (A1), so the family BH corrects is not one the proposer could have observed.

Theorem 6 (Wealth-spending FWER for verdict-adaptive proposers): For T rounds with $\mathcal{F}_{t-1}$ the past, suppose (B1) $D_{\text{hold}}^{(t)}$ is independent of $\mathcal{F}_{t-1}$, (B2) true-null p -values are super-uniform conditional on $(\mathcal{F}_{t-1}, B_t)$, (B3) within-round acceptance controls FWER at α_t (e.g. Bonferroni), and (B4) the predictable schedule satisfies $\sum_t \alpha_t \leq \alpha$. Then $\mathbb{P}(V \geq 1) \leq \alpha$ across all rounds, so $\text{FDR} \leq \alpha$.

A deployable corollary initializes $W_0 = \alpha$, fixes predictable fractions $\gamma_t \geq 0$ with $\sum_t \gamma_t \leq 1$, spends $\alpha_t = \gamma_t W_0$, and updates $W_t = W_{t-1} - \alpha_t$. Public-data caveat. Theorem 1 assumes the proposer has not seen the validation or held-out records. On public benchmarks this is threatened by pretraining contamination, so we treat public-data results as empirical stress tests rather than theorem-backed deployments. The practical safeguard we recommend is to draw the validation and held-out splits from records dated after the proposer’s training cutoff, or to keep them in a private store the proposer never sees; only then does the implementation-level firewall coincide with the independence the theorem assumes.

V. Experiments

The evaluation follows the verifier’s own decision path. We ask whether proposers produce usable constrained hypotheses (RQ1), whether gates reject plausible but invalid claims (RQ2), when the proposal layer adds value beyond statistically sound pattern mining (RQ3), and whether accepted evidence cards replicate in later data (RQ4).

Pre-specified operating choices. Thresholds are fixed before looking at held-out labels: real temporal studies use $\alpha = 0.10$ throughout. Bank and Bike use support $\tau = 0.025$ with effect $\epsilon = 0.035$ and $\epsilon = 0.050$; Healthcare uses $\tau = 0.040$ and $\epsilon = 0.010$, reflecting weaker 30-day readmission effects. Permutation tests use 500 rounds ($p_{\min} = 1/501 \approx 0.002$). Discovery, validation, and held-out partitions follow calendar time where available.

What is actually run. Although the analysis proves several properties, the deployed procedure is a single pass of Algorithm 1: freeze a batch on discovery summaries, apply BH once to the frozen batch’s validation p -values (Theorem 1), run the remaining gates, replicate on an untouched held-out split, and report cards. The weighted, multi-round, and lower-bound results (Theorems 2, 6, 4, 5) are optional or explanatory.

Baselines. LLMOnly, DescriptiveStats, StatisticalFilter, HeldoutVerifier, and FullVerifier are ablation arms of the verifier, compared in Table XIV. BH-Exhaustive enumerates singleton/pair rules and applies BH. Tarone-style, Westfall–Young maxT-style [49], and Subgroup-BH are published-style approximations of statistically

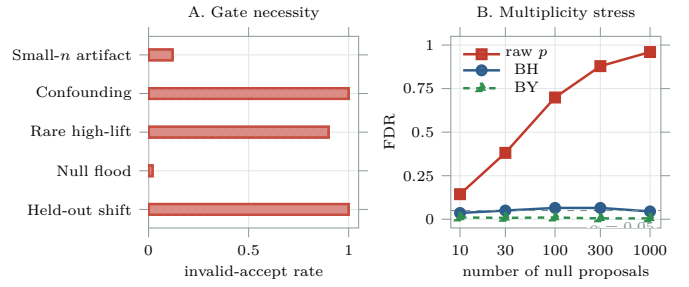


Fig. 2. Gate necessity and multiplicity stress. Left: each bar is the invalid-accept rate when the relevant gate is removed, and the full verifier accepts zero invalid claims in every regime, so its all-zero bars are omitted. Right: as null proposals grow, raw p -value testing loses FDR control while BH stays near nominal and BY is conservative.

TABLE IV
Power separation at controlled FDR.

M	Exhaustive+BH		Random-25		VeraDM	
	rec. $\uparrow$	FDR $\downarrow$	rec. $\uparrow$	FDR $\downarrow$	rec. $\uparrow$	FDR $\downarrow$
10^2	0.945	0.087	0.240	0.097	0.829	0.082
10^3	0.792	0.101	0.033	0.092	0.843	0.095
10^4	0.601	0.098	0.003	0.101	0.853	0.091
10^5	0.387	0.106	0.000	0.088	0.847	0.090

sound mining, and StableEnv-BH is an environment-stable validation/held-out variant.

A. Gate Validity and Multiplicity

Can plausible invalid claims pass the verifier? Fig. 2 (left) gives the gate-necessity matrix on a 50-seed failure-mode benchmark: each regime isolates one failure (shifted rule, null flood, rare high-lift, confounding, small- n artifact), and removing the corresponding gate admits the invalid family while the full verifier rejects it. The right panel sweeps null-proposal count: raw p -value testing reaches $\text{FDR} = 0.96$ at 1000 nulls while BH remains near nominal and BY is conservative.

B. Power Separation: the LLM Prior Has Operational Value

When does the proposer add value? Table IV evaluates Theorem 3 in an expanded temporal/sequence/treatment family. Exhaustive+BH tests the full family, Random-25+BH tests a size-matched random shortlist, and VeraDM tests 25 firewall-safe proposals. As M grows from 10^2 to 10^5 , exhaustive recall collapses from 0.945 to 0.387 under the multiplicity burden, the random shortlist drops to 0, and VeraDM keeps recall ≈ 0.85 at empirical FDR ≈ 0.09 . The Random-25 control rules out the trivial “any small family helps” explanation: the prior needs to be informative.

C. Multi-Provider Inclusion Rate and Power Separation

The power-separation result (Theorem 3) hinges on the proposer’s inclusion rate ρ , the probability that a firewall-safe shortlist contains the injected non-null hypothesis.

To test whether real LLMs clear the ρ bar in practice, we measure ρ and held-out recall at the hardest expanded-family setting $M = 10^5$ across seven proposers and three expanded-language schemas (temporal, sequence, and treatment) at “guessable” difficulty, using 50 seeds for ρ and 500 for recall. Every proposer is queried through one OpenRouter gateway with a cached call per seed, so each ρ cell rests on 50 independent proposal batches (49 in one Qwen/temporal cell, where a call returned an API error), and per-provider compile success runs from 0.89 to 1.00. Three reference arms calibrate the scale: an Oracle-25 that always includes the truth, a Human-Expert-25 hand-written shortlist, and Exhaustive+BH (Table V).

Two findings stand out. First, inclusion rate is strongly schema- and provider-dependent: Mistral-Large and Gemini-2.5-Pro clear $\rho \geq 0.70$ on multiple schemas, whereas otherwise-strong models have $\rho = 0$ on entire schemas—GPT-4o and Claude-Sonnet-4 propose no temporal family at all, and their recall collapses accordingly. This is a coverage failure rather than a parsing one: GPT-4o, for instance, compiles at rate 1.0 on the real schemas below, so its proposals are well formed but never contain the injected temporal-window hypothesis. Second, wherever ρ is high, the corresponding proposer keeps held-out recall well above Exhaustive+BH, and above the Human-Expert-25 shortlist once $\rho \geq 0.80$, matching the theorem’s prediction that recall tracks measured ρ , not model size or reputation. The practical lesson of the diagnostic is that a proposer’s per-schema ρ can be measured before it is trusted.

D. Pattern-Mining Baselines and Confounding

Is the matched-contrast gate redundant? Table VI compares VeraDM against statistically sound pattern-mining baselines on the singleton/pair family. The table uses two synthetic diagnostics that should not be conflated with the theorem-backed FDR claim: Emp. FDR counts any accepted rule that is not an injected truth as false, including redundant variants of a true signal feature, whereas the redundancy-adjusted FDR does not penalize such variants. The redundancy-adjusted column is the quantity our FDR guarantee (Theorem 1) targets, namely genuine false discoveries, and it is 0.00 for VeraDM and StableEnv-BH. VeraDM’s Emp. FDR = 0.77 here therefore reflects redundant true-signal variants rather than uncontrolled error. Validation-only methods recover true rules but accept shifted ones, whereas StableEnv-BH rejects the shifted rule by requiring environment stability. On this singleton/pair family VeraDM and StableEnv-BH coincide, so it does not by itself separate the two; the matched-contrast gate instead earns its place on the real case studies, where 16, 11, and 13 of the rejected Bank, Bike, and Healthcare cards fail it (Fig. 4). Matched contrast also drives accepted-card max-SMD to 0, and coverage pruning removes redundant accepts (Table VII). Bike is the most sensitive to this cutoff: its mean retained

overlap (0.597) is the lowest and the nearest to the 0.5 threshold. No accepted Bike card abstains at 0.5, but tightening the threshold toward 0.6 would begin to flag them, whereas Bank and Healthcare (0.925 and 0.983) stay far from it. Retained mass is therefore the diagnostic that tells a reader when to distrust a matched contrast: a value near the 0.5 floor means the contrast rests on a small matched subpopulation, whichever variables Z_h contains.

Among the arms compared here, VeraDM and StableEnv-BH are the only ones that recover true rules, refuse the shifted rule, and replicate on held-out data; VeraDM additionally admits an LLM proposer.

E. Semi-Synthetic Injection and Real Schemas

Do the gates behave on real covariates with known injected truth? We inject known true and known-null rules into real covariates over 30 seeds (Table VIII). On Adult, VeraDM recovers all injected truths with no null accepts; on BankTemporal it recovers one of three; and on German Credit none, as expected for a small dataset under held-out verification. UCI Adult Income and German Credit [50], [51] additionally serve as schema feasibility checks for the compiler.

F. Timestamped Bank Marketing Case Study

Do accepted cards replicate in later data? We treat Bank Marketing [52] as a campaign discovery problem: early months suggest targeting hypotheses, but useful discoveries should survive later campaigns. Discovery uses May–June calls ($n = 19,087$, subscription rate 7.6%), validation uses July–August ($n = 13,352$, 9.8%), and held-out replication uses November ($n = 4,101$, 10.1%). The candidate set contains 39 hypotheses, of which 14 pass all gates and 25 are rejected (Table IX, Fig. 3). Repeated small p/q values are the 500-round permutation floor, not a BH cap. Bike Sharing [53] (36 hypotheses, 22 accepted, binary target “at least 200 rentals in the hour”, with a calendar-time split into Jan–Jun 2011 discovery, Jul–Dec 2011 validation, all of 2012 held-out) and Healthcare readmission [54] (38 hypotheses, 24 accepted, encounter-order split) follow the same pattern, with per-dataset summaries in Table X and threshold sensitivity discussed below. Fig. 4 records which gate did the rejecting: no single gate dominates, and the binding constraint differs by dataset, which is the practical form of the non-redundancy claim.

Sweeping the two operating thresholds one at a time gives the expected recall/strictness tradeoff rather than a new tuning procedure. Raising support τ over $\{.01, .025, .05\}$ at fixed ϵ takes Bank from 19 to 14 to 9 accepted rules and Bike from 24 to 22 to 19; raising the effect threshold ϵ at fixed τ takes Bank from 18 ($\epsilon = .025$) to 14 (.035) to 12 (.05), and Bike from 23 to 22.

TABLE V
Full L1 multi-provider benchmark.

Model	Inclusion rate $\rho \uparrow$ [95% CI]			Recall @ $M=10^5 \uparrow$ [95% CI]		
	Temporal	Sequence	Treatment	Temporal	Sequence	Treatment
Reference arms						
Oracle-25	—	—	—	0.980 [0.971,0.986]	0.980 [0.972,0.988]	0.985 [0.979,0.992]
Human-Expert-25	—	—	—	0.716 [0.693,0.739]	0.701 [0.677,0.724]	0.702 [0.678,0.727]
Exhaustive+BH	—	—	—	0.397 [0.370,0.423]	0.388 [0.361,0.415]	0.399 [0.371,0.426]
LLM proposers (VeraDM arm)						
Mistral-Large	0.94 [0.87,1.00]	1.00 [1.00,1.00]	0.26 [0.14,0.38]	0.912 [0.898,0.927]	0.987 [0.981,0.993]	0.271 [0.248,0.294]
Gemini-2.5-Pro	0.70 [0.57,0.83]	0.86 [0.76,0.96]	0.90 [0.82,0.98]	0.671 [0.647,0.695]	0.841 [0.823,0.860]	0.876 [0.859,0.893]
Llama-3.1-70B	0.22 [0.11,0.33]	0.96 [0.91,1.00]	0.46 [0.32,0.60]	0.216 [0.196,0.236]	0.948 [0.937,0.959]	0.429 [0.404,0.453]
GPT-4o	0.00 [0.00,0.00]	0.80 [0.69,0.91]	0.18 [0.07,0.29]	0.000 [0.000,0.000]	0.780 [0.758,0.802]	0.183 [0.163,0.203]
Qwen-2.5-72B	0.04 [0.00,0.10]	0.82 [0.71,0.93]	0.04 [0.00,0.09]	0.047 [0.037,0.058]	0.811 [0.790,0.831]	0.044 [0.033,0.055]
Claude-Sonnet-4	0.00 [0.00,0.00]	0.32 [0.19,0.45]	0.00 [0.00,0.00]	0.000 [0.000,0.000]	0.319 [0.294,0.343]	0.000 [0.000,0.000]
GPT-4o-mini	0.06 [0.00,0.13]	0.12 [0.03,0.21]	0.34 [0.21,0.47]	0.061 [0.048,0.073]	0.111 [0.096,0.127]	0.337 [0.313,0.361]

TABLE VI
Statistically sound mining baselines (published-style approximations) on singleton/pair rules.

Method	Acc. $\uparrow$	Recall $\uparrow$	Emp. FDR $\downarrow$	Redund. FDR $\downarrow$	Shift $\downarrow$
BH-Exhaustive [9]	21.4	1.00	0.86	0.30	1.0
Tarone-style [22]	0.0	0.00	0.00	0.00	0.0
WY maxT-style [49]	20.7	1.00	0.85	0.29	1.0
Subgroup-BH [13]	22.2	1.00	0.86	0.32	1.0
StableEnv-BH [37]	13.0	1.00	0.77	0.00	0.0
VeraDM	13.0	1.00	0.77	0.00	0.0

TABLE VII
Balance and redundancy diagnostics.

Diagnostic	Bank	Bike	Health.
mean max-SMD before match	0.451	1.609	0.322
max-SMD after exact context	0.000	0.000	0.000
mass retained in overlap	0.925	0.597	0.983
Redundancy pruning (accepted before $\rightarrow$ after)			
Adult 17 $\rightarrow$ 14 (−3)	German 7 $\rightarrow$ 6 (−1)		

G. Multi-Round Adaptive Discovery and LLM Proposer Diagnostics

We split each Bank and Bike held-out period into three fresh confirmatory shards and run a verdict-adaptive proposer with predictable schedule $(\alpha_1, \alpha_2, \alpha_3) = (0.05, 0.03, 0.02)$. Within each round, acceptance uses Bonferroni on the fresh shard plus the standard gates. Bank accepts 0/0/4 duration-based claims across rounds, and Bike accepts 6/3/5. Prior verdicts may shape later prompts, but no round reuses a confirmatory shard after its evidence card is revealed.

The constrained grammar makes proposal failures visible rather than mysterious. API-model proposals compile reliably: gpt-4o-mini and the stronger gpt-4o each generate 24–25 unique hypotheses per dataset at compile success 1.0. Local open-weight small language models (SLMs) are weaker stress tests: smollm2:135m produces no parseable hypotheses, llama3.2:3b compiles at 0.33, qwen2.5:0.5b at 0.75 (2.0 unique), and llama3.2:1b reaches 0.80 (4.8 unique). Because the verifier is unchanged across pro-

TABLE VIII
Semi-synthetic injected-rule benchmark on real covariates, 30 seeds.

Dataset	Recall $\uparrow$	Emp. FDR $\downarrow$	Accepted
Adult	1.00	0.00	3.0
German	0.00	0.00	0.0
BankTemporal	0.33	0.00	1.0

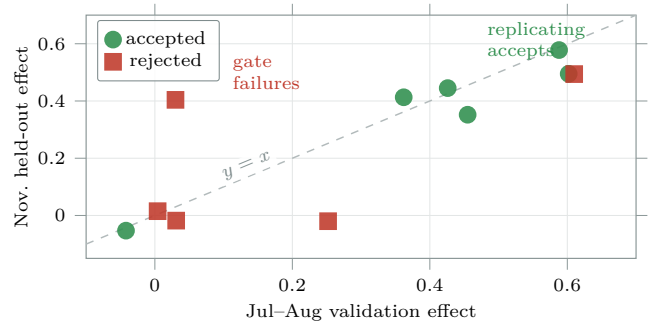


Fig. 3. Selected Bank evidence cards; accepted cards cluster near the $y=x$ replication line.

posers, poor proposals are absorbed at the compiler or evidence-card stage. On the real schemas the accepted rules are mostly within the enumerable singleton/pair space with a cross-family minority, verified at 9.5–22.7 hypotheses per second, and the grammar compiles every valid expanded-language kind while rejecting every invalid one (Table XI). Confidence-weighted BH (Theorem 2) is a diagnostic rather than a deployable gain here: oracle weights built from held-out outcomes recover full power, which upper-bounds what any real confidence signal could achieve, while miscalibrated weights that reuse the oracle level inflate FDR (Table XII).

H. Auditability: Firewall Leakage Audit

Does the firewall hold in practice? The FDR result (Theorem 1) rests on the proposer not seeing validation or held-out data. We audited all 89 proposer prompts across the case studies: every prompt contained the schema and discovery summaries (100%), and none included any

TABLE IX
Selected Bank Marketing evidence cards.

Decision	Rule	Sup.	Val.	Hold.	Reason
acc.	duration300 $\uparrow$ AND euribor $\uparrow$	.04	.59	.58	passes
acc.	recent prior contact $\uparrow$	.03	.60	.50	passes
acc.	euribor low $\uparrow$	.11	.36	.41	passes
acc.	duration $\uparrow \geq 600$	.09	.46	.35	passes
acc.	blue collar $\downarrow$ AND campaign $\downarrow$	.07	-.04	-.05	passes
rej.	poutcome success $\uparrow$	.02	.61	.49	support
rej.	confidence $\uparrow \geq -40$	.48	.03	.40	weak contrast
rej.	cellular $\uparrow$	.90	.03	-.02	no replication
rej.	university $\uparrow$	.35	.00	.02	FDR

TABLE X
Bike and Healthcare case-study summaries.

Dataset	Hyp.	BH acc.	BY surv.	max-SMD after $\downarrow$	overlap kept $\uparrow$
Bank	39	14	14	0.000	0.925
Bike	36	22	22	0.000	0.597
Healthcare	38	24	24	0.000	0.983

validation or held-out record (leak rate 0.000). The firewall is thus satisfied operationally, not only by construction. This audit is implementation-level: it certifies that no validation or held-out record entered a prompt at run time, but it cannot certify statistical independence from a public dataset the proposer may have seen during pretraining (Sec. IV). Under label randomization (Bank, 50 seeds with permuted targets) the verifier accepts 0 rules on every seed (Table XIII), a direct negative control for an FDR-control claim. The compiler itself passes an 80-item gold set (60 valid, 20 invalid) at compile rate 1.00, predicate F1 1.00, and direction accuracy 1.00. It fails closed in the strict sense of rejecting any claim that does not type-check against the schema, and the 0.80 invalid-rejection rate reflects that the remaining gold-set items are well typed but semantically invalid, so they compile and are instead caught downstream by the statistical gates, which leaves firewall safety unaffected.

I. Verifier Arms

What does each layer of the gate stack buy? Table XIV runs the five arms over identical proposals, so they differ only in which gates are enabled. Every arm recovers every injected pattern, so what separates them is what they wrongly admit: the unfiltered and validation-only arms accept the spurious pattern on all 24 seeds, and the spurious-accept rate falls to zero exactly when held-out replication is switched on. Matched contrast changes nothing here, so its value is visible only on the real case studies (Fig. 4).

J. Cost and Reproducibility

Verification is inexpensive: throughput is inversely proportional to the permutation budget R (Table XI), a full Bank or Bike run completes in seconds on one CPU core with no GPU, and proposal is a single batched LLM call.

Reproducibility. All thresholds are fixed before any held-out split is read. Seed counts are per experiment: 50 for the failure-mode matrix, 200 for the null sweep, 500 for power separation, 30 for the semi-synthetic injections, and 10 for

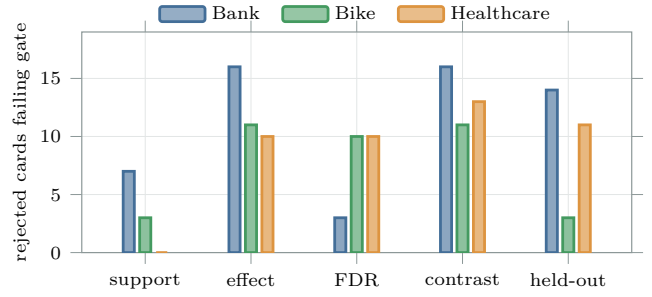


Fig. 4. Which gate does the rejecting, over the 25/14/14 rejected cards on Bank/Bike/Healthcare; a card may fail several gates. Support binds on Bank but never on Healthcare, where contrast and held-out replication carry the rejections.

TABLE XI
Novelty, throughput, and language coverage. Throughput is profiled on the two clean timestamped datasets (Bank and Bike).

	Bank	Bike	Health.
accepted / enumerable	14/14	22/21	24/24
cross-family / expanded-lang.	5/0	7/1	5/0
hyp./s @ 200/500/1000 rounds	19/9/5	40/23/14	—
Language coverage: 4 kinds, valid-compile 1.00, invalid-rejection 1.00, 5/5 expanded items verified.			

the mining baselines; permutation tests use 500 rounds. The public datasets are UCI Adult, German Credit, Bank Marketing, Bike Sharing, and Diabetes 130-US hospitals [50], [51], [52], [53], [54]. Verification runs CPU-only. Code, cached proposer batches, and per-run logs are available at <https://github.com/LeoTTTPhat/VeraDM>.

VI. Discussion

Our experiments support the design principle introduced at the start: in LLM-guided data mining, proposal and acceptance need different mechanisms. An LLM is a useful source of interpretable candidate rules, but fluency alone does not tell us whether a claim holds. A plausible claim can be confounded, under-supported, a multiplicity artifact, or unstable in later data, so VeraDM treats the model as a candidate generator and the verifier as the authority on what counts as a discovery.

The same separation makes the FDR result usable in practice, with two operating modes. In the default batch mode, control is conditional on a frozen proposed set tested against splits the proposer never saw (Theorem 1). The moment the proposer is allowed to see verdicts and react, a single reused held-out set is no longer enough (Theorem 4), and the interactive mode restores control by spending a predictable wealth schedule against fresh confirmatory shards (Theorem 6). Intuitively, letting the model re-query a reused held-out set after seeing verdicts is like re-rolling a test until it passes, and the wealth schedule simply caps the total significance budget those re-rolls may spend.

Two results explain where the LLM layer earns its place. Power separation (Theorem 3) shows that the prior's

TABLE XII

Confidence-weighted BH on Bank temporal. Oracle weights are built from held-out outcomes and upper-bound achievable power.

Method	Disc.	Emp. FDR↓	Power↑
Uniform BH	0	0.00	0.00
Weighted BH (oracle)	21	0.00	1.00
Weighted BH (miscalibrated)	0	0.00	0.00
miscalibrated @ oracle α	10	1.00	0.00

TABLE XIII

Audit dashboard for assumptions and reporting.

Audit	Main result
firewall prompt scan	89 prompts audited, 0 held-out leaks
label randomization	0 accepts over 50 Bank seeds
BY dependence check	survival 1.00 on all three (min q 0.011–0.014)
verification throughput	Bank 9.49 hyp./s, Bike 22.68 hyp./s

job is to shrink the tested family from a vast semantic space to a small firewall-safe batch without invalidating BH, and the case studies show the matched-contrast gate doing independent work: 16, 11, and 13 of the rejected Bank, Bike, and Healthcare cards fail it. The proposer plays two roles depending on the search space. On the timestamped real schemas, where accepted rules mostly lie in the enumerable singleton/pair space, its value is budget efficiency. It recovers the same discoveries from a batch of a few dozen proposals instead of exhaustive search, together with 5 to 7 cross-family conjunctions per dataset that singleton/pair enumeration would not surface as a unit (Table XI). On the expanded-language L1 families (M up to 10^5), the effect is larger: exhaustive BH collapses under multiplicity, while a high-inclusion-rate proposer preserves held-out power (Table V).

Finally, auditability here is more than a formatting choice. The prompt audit finds zero held-out leakage, label randomization accepts nothing over 50 seeds, and every real-data accept survives Benjamini–Yekutieli under arbitrary dependence.

A. Practical Deployment and Cost

VeraDM is inexpensive to run and easy to drop in. Proposal is a batched LLM call (about 25 hypotheses per call) and verification is one pass with no fine-tuning: throughput is 9.5–22.7 hypotheses/s at 500 permutations and falls inversely with the permutation budget, which simply trades runtime for a lower p -value floor (Table XI). Because the verifier is identical across proposers, VeraDM integrates as a thin wrapper around an existing model or mining pipeline rather than a replacement for it. The operator’s main lever is the support/effect threshold, and the number of discoveries responds to it smoothly and predictably, so the recall/strictness tradeoff can be dialed without re-engineering the gates.

TABLE XIV

Verifier arms (24 seeds, 24 proposals per seed). Spurious accept is the fraction of seeds accepting the injected spurious pattern; as in Table VI, Emp. FDR counts redundant variants of a true pattern as false.

Arm	Acc.	Recall↑	Emp. FDR↓	Spurious↓
LLMOnly	23.0	1.00	0.87	1.00
DescriptiveStats	15.3	1.00	0.80	1.00
StatisticalFilter	15.2	1.00	0.80	1.00
HeldoutVerifier	9.0	1.00	0.67	0.00
Full stack (VeraDM)	9.0	1.00	0.67	0.00

B. Limitations

The compiler targets binary outcomes and a limited rule language, and it fails closed on unknown features and missing directions but not on malformed comparisons: an unrecognized operator is dropped rather than rejected, leaving a predicate that tests the feature for presence. Matched contrast uses exact-match strata, and propensity weighting and doubly robust estimation are natural extensions. The Healthcare split is an encounter-order proxy, so Bank and Bike remain the clean timestamped evaluations. Our splitting and permutation arguments also treat rows as exchangeable observational units: a Bank row is one contact, a Bike row one calendar record, and a Healthcare row one encounter, so calendar-ordered serial dependence and repeated patients across encounters are not modeled, and block- or cluster-aware permutation is the natural extension. Baselines include exact exhaustive BH and published-style corrections, not bundled reference implementations of every prior system. The LLM proposer study uses cached API and local SLM batches rather than a full temperature/prompt/scale sweep. Finally, the FDR result invokes PRDS for its BH step. This is plausible but not verified for heavily overlapping rule families, so in practice we treat BY survival (Table XIII) as the operative robustness check and recommend BY whenever the proposed batch is strongly co-dependent. Every dataset here is public, so the held-out guarantee carries the pretraining-contamination caveat of Sec. IV, and the firewall targets the non-public, proprietary setting where that risk is absent, which makes an at-scale proprietary deployment the priority next step. Evidence cards are designed for auditability, but we do not yet have a masked user study establishing that they improve human judgment; a pre-registered study with the verifier’s verdict hidden is planned.

VII. Conclusion

We studied LLM-guided data mining under a simple discipline: an LLM may propose constrained natural-language hypotheses, but discoveries should be accepted only after evidence checks. VeraDM turns this discipline into a working system: a firewall-preserving compiler, a multi-gate verifier with held-out replication and matched contrast, evidence cards as an auditable reporting unit, and a power-separation result that characterizes when

the proposal layer helps. The broader lesson is the one we want readers to carry forward: an LLM is valuable as a hypothesis generator, not as a discovery authority. Credibility should rest on typed queries, controlled testing, held-out replication, and transparent reporting, not on how convincing a claim sounds. We hope the verifiable-discovery framing makes that separation easier to demand in LLM-assisted analysis.

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